

HOUSTON SOUTHWEST, TX, USA

WMO: 720637

Lat: 29.500N

Lon: 95.477W

Elev: 69

StdP: 14.66

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00223

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.9	33.1	14.2	11.3	40.5	19.7	14.8	43.1	21.0	52.4	18.8	53.0	3.8	350	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.8	96.8	75.4	94.8	76.3	92.9	76.6	80.7	87.2	80.0	87.2	79.3	87.0	6.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.1	82.3	78.2	146.8	82.1	77.3	142.5	81.9	44.2	87.2	43.4	87.9	42.7	87.0	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.3	12.8	24.1	99.1	4.2	2.7	21.1	101.0	18.6	102.6	16.2	104.1	13.2	106.0		
(5)			22.2	81.6	4.4	0.9	19.0	82.2	16.4	82.7	13.9	83.2	10.6	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	52.2	57.8	63.6	69.2	75.5	81.1	82.2	83.3	79.1	70.5	61.1	55.0		
	DBStd	12.72	9.35	9.93	7.88	6.30	5.18	2.99	2.83	2.45	3.73	7.18	9.40	9.33		
	HDD50	197	84	38	7	0	0	0	0	0	0	0	17	51		
	HDD65	1353	405	237	123	33	5	0	0	0	0	31	183	336		
	CDD50	7233	154	256	429	576	791	932	998	1032	872	637	350	206		
	CDD65	2915	10	36	80	159	331	482	533	567	422	202	67	26		
	CDH74	27329	37	138	378	1017	2865	4995	5661	6156	3749	1818	401	114		
	CDH80	11705	0	9	42	169	970	2365	2724	3137	1588	621	74	6		
(14) Wind	WSAvg	4.9	5.8	6.3	6.1	6.2	5.5	4.0	3.1	3.3	3.5	4.3	4.9	5.7		
(15) Precipitation	PrecAvg	53.2	4.3	3.0	3.7	4.1	5.2	5.8	4.5	4.3	4.7	5.4	4.2	4.1		
	PrecMax	79.5	10.5	9.1	8.5	9.8	13.2	24.6	12.1	9.1	11.5	17.6	12.7	8.3		
	PrecMin	23.7	0.6	0.5	0.8	0.1	0.0	0.3	1.4	0.5	1.6	0.0	0.7	1.1		
	PrecStd	13.0	2.4	2.2	2.1	3.0	3.3	4.8	2.9	2.6	2.3	4.5	2.8	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.7	81.6	83.5	87.3	92.7	97.8	97.4	99.4	95.5	90.8	85.9	80.8		
		MCWB	66.1	68.8	68.6	72.4	74.0	74.1	75.3	75.0	74.4	74.5	71.8	68.4		
	2%	DB	74.6	78.3	81.0	84.0	90.0	95.0	95.3	97.2	92.9	88.7	82.0	77.4		
		MCWB	64.6	67.6	68.8	71.1	74.2	75.5	76.3	76.1	75.5	75.2	70.9	69.2		
	5%	DB	71.4	75.1	78.5	81.8	87.9	92.5	93.4	95.0	90.7	86.4	78.8	73.8		
		MCWB	63.8	67.0	68.0	70.3	74.0	76.3	76.9	77.0	75.5	74.2	69.8	68.6		
	10%	DB	68.1	72.0	75.5	79.8	85.8	90.6	91.4	93.0	88.5	83.8	75.2	71.4		
		MCWB	62.7	66.3	66.4	69.2	73.7	76.6	77.2	77.3	75.7	72.3	68.2	66.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.5	71.9	72.8	77.1	78.5	81.0	81.1	81.7	81.0	83.9	76.1	73.4		
		MCDB	72.3	76.7	78.8	82.5	85.6	88.0	86.8	88.1	84.3	85.1	81.6	75.4		
	2%	WB	68.6	70.4	71.5	75.1	77.5	79.9	80.2	80.7	79.5	78.2	74.1	71.7		
		MCDB	71.4	74.6	77.0	80.0	84.0	87.2	87.3	88.3	85.5	84.6	78.1	74.7		
	5%	WB	66.8	69.4	70.5	73.7	76.6	79.0	79.5	80.0	78.6	76.7	72.7	70.2		
		MCDB	70.4	72.5	75.0	78.3	83.2	86.5	87.6	88.0	85.1	83.1	76.1	73.4		
	10%	WB	63.7	68.1	69.5	72.5	75.5	78.2	78.8	79.2	77.8	74.9	70.8	67.8		
		MCDB	66.2	71.1	73.0	76.8	82.3	85.9	86.7	87.4	84.6	80.5	74.5	70.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.5	18.5	19.2	19.5	19.0	18.9	19.0	19.8	18.1	21.9	21.2	18.8		
		MCDBR	23.3	19.6	19.8	19.8	20.4	21.9	22.0	23.1	21.6	22.3	21.7	21.5		
		MCWBR	15.2	11.0	10.0	9.7	7.1	6.7	6.9	6.9	7.6	10.2	12.0	14.7		
	5% WB	MCDBR	18.1	16.2	16.0	15.2	16.2	18.1	19.0	19.9	17.0	19.0	17.6	19.7		
		MCWBR	14.6	11.2	9.6	9.0	6.7	7.1	7.0	7.2	7.0	10.3	11.4	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.360	0.376	0.438	0.482	0.469	0.496	0.484	0.455	0.381	0.374	0.352		
	taud		2.491	2.456	2.402	2.229	2.141	2.233	2.174	2.210	2.258	2.457	2.424	2.475		
	Ebn at Noon		283	287	289	273	260	262	254	257	261	277	269	273		
	Edh at Noon		29	33	37	45	49	45	48	45	42	32	31	28		
(44) All-Sky Solar Radiation	RadAvg		913	1030	1407	1675	1866	1929	1878	1808	1570	1377	1038	813		
	RadStd		88	177	119	124	136	158	186	153	169	161	105	98		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+203	+178

Nomenclature: See separate page